

In a distributed system, these non-functional characteristics have specific meanings.

Consistency: All data across all nodes in a distributed system is the same.

Availability: Every request made to the database receives a response, even if some nodes are unreachable.

Partition tolerance: The system continues to operate even if network partitions occur, adversely affecting nodes and creating isolations.

The CAP theorem provides a valuable framework for screening databases when designing distributed systems. It is named after all the three critical DBMS properties — consistency, availability and partition tolerance. According to the theorem, a distributed system must choose between consistency and availability depending on the presence of network partitions. This means that under network partitions, the system can either sacrifice consistency to maintain availability or sacrifice availability to maintain consistency.

While the CAP theorem offers insights into current distributed systems, in practical implementations, availability and consistency can be adjusted and configured to meet specific requirements. Ensuring adherence to each property requires the skills of seasoned database architects in designing, as well as subject matter experts who ensure that configurations

are updated to achieve the desired database performance.

Open source databases, like any other distributed databases, are also subject to the CAP theorem. The design choices made by the developers of these databases determine whether they prioritise consistency or availability in the face of network partitions.

Let's take a look at some of the popular open source databases, each with its own approach to the CAP theorem.

Cassandra, developed by Apache foundation, is a NoSQL database that emphasises availability and partition tolerance over strong consistency. It provides tunable consistency levels, allowing users to choose their desired trade-off between consistency and availability.

MongoDB is another NoSQL database that prioritises availability and partition tolerance. It offers eventual consistency by default, but also provides options for stronger consistency guarantees if required.

CockroachDB is a distributed SQL database that aims to provide strong consistency, high availability, and partition tolerance. It uses a distributed consensus protocol called Raft to ensure consistency across its nodes.

MySQL is a relational database system (RDBMS) designed to provide atomicity, consistency, isolation, and durability (ACID).

YugaByteDB is a distributed SQL database that aims to provide strong consistency, high availability, and partition tolerance. It adopts CP (consistency-partition tolerance), prioritising consistency and partition tolerance in the event of network partitions, sacrificing immediate availability.

Neo4J is a graph database primarily deployed as a single node or small scale clustered database. However, when deployed in distributed systems, it adheres to CA with strong constancy and transactional integrity.

Redis is another database which is deployed in a single-server model in most use cases. However, when deployed in distributed systems, it adheres to CP, lacking availability which needs to be addressed by deployment architecture.

HBase, developed by the Apache foundation, is a columnar database designed for consistency and partitioning in terms of workload profile management.

PostgreSQL is the most popular open source relational database and continues to be the leading choice across all types of organisations. It adheres to CA, leading to implementation focus on partition tolerance.

Any of these databases can be configured for any of the three properties. Striking the right balance between the properties would depend on the use case.

Table 1 provides an overview of the characteristics and properties

Database	Data model	Transaction model	CAP theorem adherence
MySQL	Relational	ACID	Consistency – Availability
PostgreSQL	Relational	ACID	Consistency – Availability
Neo4J	Graphical	ACID	Consistency – Availability
Cassandra	Column oriented	BASE	Availability – Partition Tolerance
CouchDB	Document oriented	BASE	Availability – Partition Tolerance
YugabyteDB	Relational	BASE	Availability – Partition Tolerance
Redis	Key-value	ACID	Consistency – Partition Tolerance
Mongo	Document oriented	ACID	Consistency – Partition Tolerance
HBase	Column oriented	ACID	Consistency – Partition Tolerance

Table 1: Characteristics and properties of open source databases